

Product Kernel Perspective Spaces: Predicting Benchmark Scores from Sparse, Unpaired Model Evaluations

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Abstract

Evaluating language models is expensive, and in practice benchmark coverage is sparse: many models, many tasks, but each model is run on only a subset of tasks, and within a task on only a subset of queries. The Data Kernel Perspective Space (DKPS) embeds models from their responses and predicts held-out scores, but it requires a *paired* design—every model must answer the same queries—so it discards the abundant unpaired evaluations real benchmarks produce. We introduce the **Product Kernel Perspective Space (PKPS)**, which replaces DKPS’s implicit identity matching of queries with a *product kernel* $k_Q \cdot k_R$ that weights response comparisons by query similarity. PKPS recovers DKPS exactly in the paired limit (as the query-kernel bandwidth $\sigma \rightarrow 0$) and degrades gracefully when models answer different queries. On synthetic data we map how prediction accuracy depends on the number of models, task sparsity, query sparsity, and task structure. On the HELM benchmark we study two regimes with their domain-standard baselines: (i) *missing queries* (efficient benchmarking), where PKPS matches item-response theory (IRT) when data is paired but, unlike IRT and DKPS, does not collapse as queries unpair; and (ii) *missing tasks* (benchmark completion), where PKPS is competitive with low-rank matrix completion and wins decisively when few models are available. We then scale to a single heterogeneous suite of 18 tasks and 93 models spanning math, translation, medical QA, and legal classification, using a block-diagonal product kernel that places rich-text and single-token responses in one space: PKPS roughly halves the error of naive sampling for efficient evaluation across the suite. In every regime an ensemble of PKPS with the domain baseline is the most robust predictor, indicating the embedding and score-only signals are complementary.

1 Introduction

Comprehensive evaluation of language models is a growing bottleneck: leaderboards track hundreds of models across dozens of benchmarks, yet no model is run on everything. The resulting (model $\times$ task) score matrix is *doubly sparse*. *Across tasks*, a model is evaluated on a subset of benchmarks (missing tasks). *Within a task*, evaluation may use only a subset of queries to save cost (missing queries). Predicting the missing entries—a model’s score on a task it has not been (fully) run on—would let practitioners reuse cached evaluations instead of paying to re-run them [Polo et al., 2024, Vivek et al., 2024, Papailiopoulos et al., 2026].

The Data Kernel Perspective Space (DKPS) [Anonymous, 2026] is a black-box approach: it embeds each model from its query responses, places models in a low-dimensional Euclidean space via classical multidimensional scaling, and regresses benchmark scores in that space. DKPS is query-efficient and competitive with item-response theory. But it has a structural limitation: the distance between two models is the average squared difference of their responses *to the same queries*, so DKPS requires a *paired* design. When models answer different queries—the common case in real, organically-assembled leaderboards—DKPS has nothing to compare and the information in those unpaired responses is thrown away.

We introduce the **Product Kernel Perspective Space (PKPS)**. The key change is to compare responses to *similar* queries, not only identical ones, by weighting each response-pair comparison with a query kernel. This (i) lets every unpaired response contribute, and (ii) contains DKPS as the special case where the query kernel is the identity. Our contributions:

- **Method.** A product-kernel distance $k_Q \cdot k_R$ that generalizes DKPS, reduces to it as the query-kernel bandwidth $\sigma \rightarrow 0$, and whose bandwidth is chosen per run by a cheap median-scaled cross-validation (Section 2).
- **Synthetic study** of how accuracy scales with the number of models, task sparsity, query sparsity, and task spread (Section 3).
- **Missing queries (RD1).** Using HELM-MATH as a running example, PKPS recovers the query-efficiency of DKPS in the paired regime and—unlike DKPS and IRT, which both collapse—remains accurate as queries unpair (Section 4).
- **Missing tasks (RD2).** PKPS is competitive with low-rank matrix completion, wins when models are scarce, and an ensemble of the two is best, showing the embedding and low-rank signals are complementary (Section 5).
- **Joint heterogeneous suite.** Our main result evaluates both regimes on a single 18-task, 93-model benchmark spanning math reasoning, translation, medical QA, and legal classification (Section 6). A block-diagonal product kernel handles datasets with rich text responses and with single-token answers in one space. PKPS roughly halves the cost of efficient evaluation across the suite; for benchmark completion the suite matrix is low-rank and matrix completion is strong, but the PKPS ensemble remains the most robust predictor.

2 Background and method

Paired DKPS. Let $x_{ij} = \text{embed}(\text{model}_i(q_j))$ be the embedding of model i ’s response to query j . With all n models answering the same m queries, the DKPS squared distance is $D^2(i, k) = \frac{1}{m} \sum_j \|x_{ij} - x_{kj}\|^2 = A_{ii} + A_{kk} - 2A_{ik}$, where $A_{ab} = \frac{1}{m} \sum_j k_R(x_{aj}, x_{bj})$ and k_R is a linear response kernel. Classical MDS on D yields model coordinates; a regressor maps coordinates to benchmark scores. The sum over j pairs response j of model a with response j of model b : an *identity* matching of queries.

Product kernel. PKPS replaces identity matching with a query kernel k_Q :

$$A_{ab} = \frac{\sum_{j \in Q_a} \sum_{l \in Q_b} k_Q(q_j, q_l) k_R(x_{aj}, x_{bl})}{\sum_{j \in Q_a} \sum_{l \in Q_b} k_Q(q_j, q_l)}, \quad (1)$$

where Q_a is the set of queries answered by model a and q_j is the embedding of query j . Each term is normalized by its own query-kernel mass, so models that answered different numbers of (or entirely different) queries are still comparable. Two limits are important:

- $k_Q = \delta$ (**identity**): only $j = l$ survives, recovering paired DKPS.
- k_Q **RBF**: $k_Q(q_j, q_l) = \exp(-\|q_j - q_l\|^2 / 2\sigma^2)$ lets responses to *similar* queries—even across tasks—contribute, so unpaired data is used.

As $\sigma \rightarrow 0$ the RBF kernel converges to δ (distinct queries have distinct embeddings), so PKPS is a strict generalization of DKPS with DKPS as its $\sigma \rightarrow 0$ limit.

Choosing the bandwidth. The optimal σ depends on how paired the data is: small when query overlap is high (recover exact matching), large when overlap is low (bridge across queries). We set it per run by cross-validation over a median-scaled grid $\sigma \in \{0.02, \dots, 5\} \cdot \text{med}$, where med is the median pairwise query distance. The selection signal differs by regime, and this dictates how aggressively we tune. In the *missing-queries* setting the reference models’ full scores are known and dense, so leave-one-family-out gives a low-noise criterion and the grid search is reliable. In the

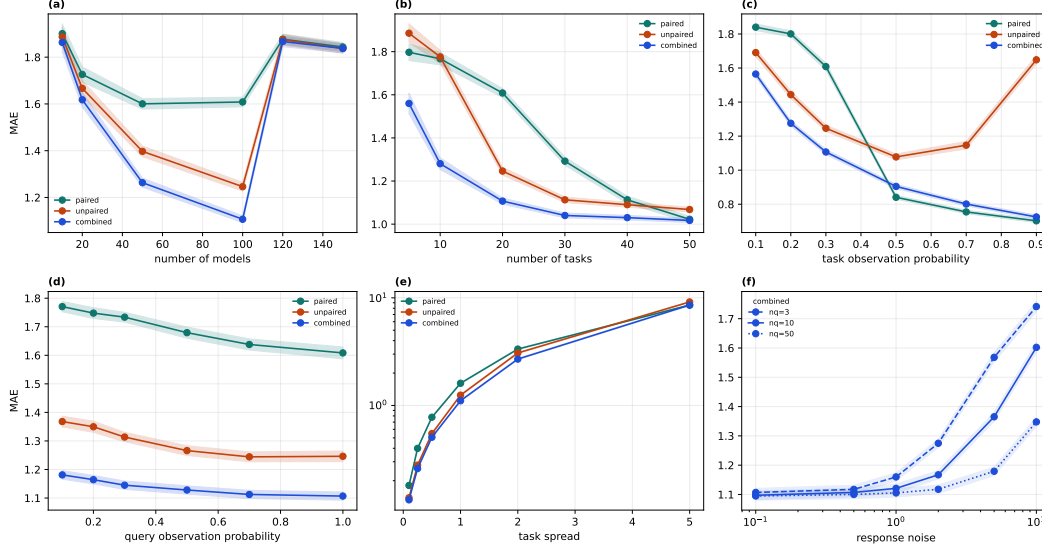


Figure 1: Synthetic study (held-out score MAE, 50 seeds). The combined product kernel dominates the paired/unpaired estimators across the number of models, number of tasks, task observation probability, query observation probability, task spread, and response noise $\times$ queries-per-task.

missing-tasks setting only a sparse set of held-out cells is available; there we find held-out MAE is flat in σ for $\sigma \geq 0.5$ med, so a fixed median bandwidth is both simpler and more robust than per-run selection on the noisy split. One response-kernel pass produces the entire σ -grid, so tuning is nearly free. We use a single regressor (distance-weighted k -NN) and a logit-space, bias-decomposed head throughout; the one quantity we always fit per run is the ensemble weight that blends PKPS with the domain baseline.

Estimation pipeline. For each replicate we (i) reduce response and query embeddings by PCA on the *observed* rows only (dimension chosen by the second Zhu–Ghodsi elbow); (ii) form D from Eq. 1; (iii) embed models by classical MDS; (iv) predict scores with a bias-decomposed, logit-space head—a two-way (model + task) additive model plus a regression of the residual on the embedding—matching the normalization used by strong score-only baselines.

3 Synthetic experiments

We generate models with latent ability vectors v_i and tasks with latent vectors t_k , scores $v_i \cdot t_k + \epsilon$, and queries drawn near each task center. Two-level missingness (task-level and query-level) mirrors real benchmarks. Figure 1 sweeps, one at a time, the number of models, number of tasks, task observation probability, query observation probability, task spread, and response noise. The *combined* estimator (all queries) consistently beats the *paired* (shared-task only) and *unpaired* (unique-task only) estimators, with the largest margin in the realistic regime: many models, sparse task overlap, moderate task spread. The advantage shrinks when overlap is abundant (paired already has the data it needs) or tasks are very dissimilar (the query kernel cannot bridge them).

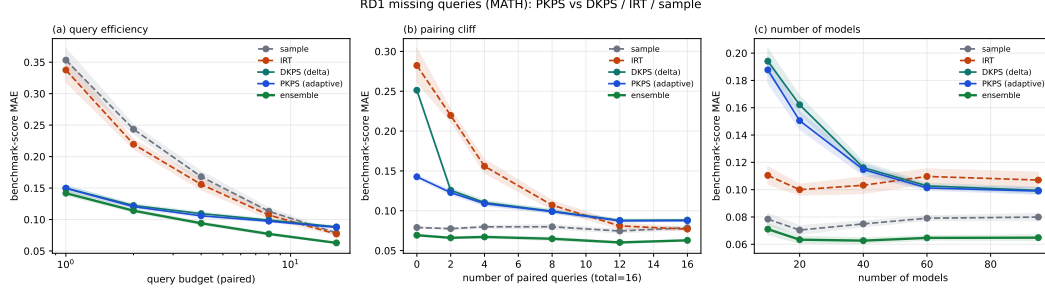


Figure 2: Missing queries on HELM-MATH (benchmark-score MAE, 20 seeds). (a) Query efficiency: DKPS/PKPS beat sample and IRT at low budget. (b) Pairing cliff: IRT and DKPS collapse as queries unpair; PKPS survives. (c) Number of models: DKPS collapses with few models, PKPS is robust. The ensemble is best throughout.

4 Missing queries: efficient benchmarking

Here each model is run on a few queries of a single benchmark and we predict its *full* benchmark score, leave-one-model-family-out. Baselines: the *sample* mean of answered queries; *IRT* (1-PL Rasch), which fits item difficulties on the largest fully-paired $|F| \times |Q|$ block of the observed responses and predicts ability from the answered anchor items; and *DKPS* (identity query kernel). We report PKPS with the adaptive bandwidth and an ensemble of PKPS with the sample mean (mixing weight fit per run). All use Gemini response embeddings on 95 models / 39 families.

Figure 2 shows three regimes. **(a) Query efficiency** (all models answer the same m queries): at a budget of one query, DKPS/PKPS reach 0.14 MAE versus 0.35 for the sample mean and 0.34 for IRT; the sample mean catches up only as the budget grows. **(b) The pairing cliff** (fix the budget at 16 queries, slide from fully paired to fully unpaired): IRT and DKPS are strong when paired (0.077, 0.087) but *collapse* as queries unpair (0.28, 0.24), because neither has a dense block or shared queries to use; PKPS degrades only mildly (0.09 $\rightarrow$ 0.14). **(c) Number of models**: both perspective methods need a cohort to regress against and degrade together when models are scarce (~ 0.19 at $n=10$), converging as the cohort grows. The ensemble is best in every panel: when the structured methods fail, it falls back to the sample mean; when queries are few, it leans on the embedding.

The takeaway: in the *paired* regime PKPS reproduces the query-efficiency of DKPS and IRT, but only PKPS retains accuracy once evaluations are *unpaired* or models are scarce—the regimes that dominate real, organically-collected leaderboards.

5 Missing tasks: benchmark completion

Now we predict held-out (model, task) entries of a (model $\times$ task) matrix. The domain-standard baseline is low-rank matrix completion: model-and-task biases plus a rank-2 factor fit by alternating least squares in logit space [Papailiopoulos et al., 2026]. We compare PKPS and an ensemble of the two, with the blend weight fit per run by repeated held-out cross-validation. Each observed pair contributes its responses to the embedding and its (sample) score to the completion; held-out MAE is measured against the true full score.

On MATH (Figure 3), PKPS beats matrix completion at every task-observation probability, and as the number of models shrinks matrix completion collapses (rank-2 completion needs many rows) while PKPS is flat— $2\times$ lower at $n=10$. An *ensemble* of the two is best at essentially every operating point: the embedding and low-rank signals are complementary, and the cross-validated weight defaults to the stronger component when the cohort is too small to fit a blend. On WMT-14, whose

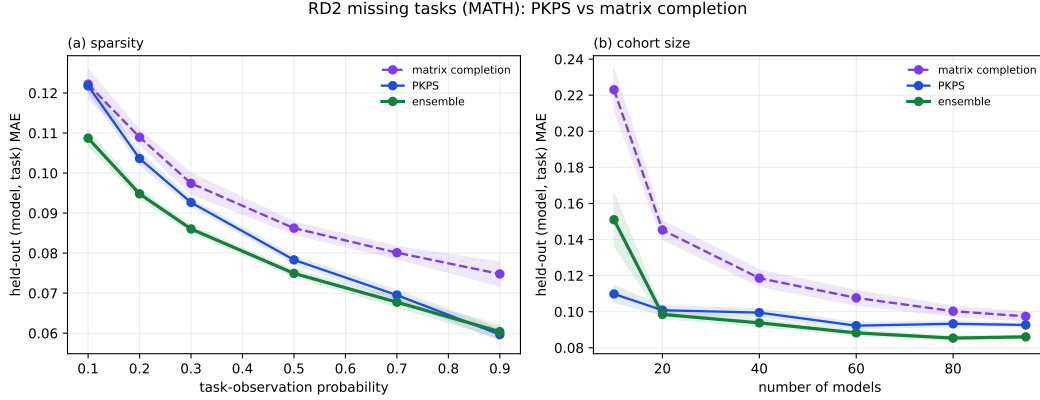


Figure 3: Missing tasks on HELM-MATH (held-out MAE). PKPS beats matrix completion at every task-observation probability (left); at every model count (right) matrix completion collapses when models are scarce—rank-2 completion needs many rows— while PKPS stays flat ($2\times$ lower at $n=10$). The ensemble (green) is best throughout, falling back to whichever component is stronger when the cohort is too small to fit the blend.

continuous translation scores form a near-rank-one matrix, the picture inverts: matrix completion is the stronger method and PKPS trails, but the ensemble stays competitive—the two datasets bracket the spectrum of matrix rank. A per-task / per-model error breakdown (Figure 4) shows the ensemble has the lowest error on all seven MATH subjects and shifts the entire per-model error distribution down, rather than helping a few outliers.

6 A joint heterogeneous suite

The single-dataset studies use MATH as a running example to build intuition. Our main result puts both regimes on one realistic, organically-assembled benchmark: a suite of 18 tasks across four datasets with very different response types—**MATH** (7 math-reasoning subjects), **WMT-14** (5 translation directions), **MedQA** (medical multiple-choice), and **LegalBench** (5 legal classification subsets)—restricted to the 93 models evaluated on all four.

A block-diagonal product kernel. MATH and WMT produce rich free-text responses (worked solutions, translations) whose Gemini embeddings carry signal, but MedQA and LegalBench responses are single tokens (a choice letter, a class label) for which text embeddings are degenerate. We therefore represent each dataset’s responses natively—Gemini embeddings for the free-text datasets, one-hot answer encodings for the single-token ones (so a linear k_R measures answer agreement)—and place them in *disjoint blocks* of the response space. With a linear response kernel this makes k_R exactly zero across datasets, while a single shared query embedding with a *within-domain* bandwidth keeps k_Q near zero across domains: the product kernel automatically factorizes by domain, bridging similar queries within a dataset and never forcing a spurious comparison between a proof and a legal label. The rest of the pipeline is unchanged.

Efficient evaluation (missing queries). Each model answers only m queries per task; we estimate every (model, task) full score (Figure 5). Because all tasks are observed—only queries are missing—matrix completion does not apply, and this is PKPS’s regime. The PKPS + sample ensemble dominates naive sampling at every budget and on all four datasets, roughly *halving* the error at small budgets (suite-wide MAE 0.13 vs. 0.29 at one query per task; 0.093 vs. 0.135 at four). PKPS

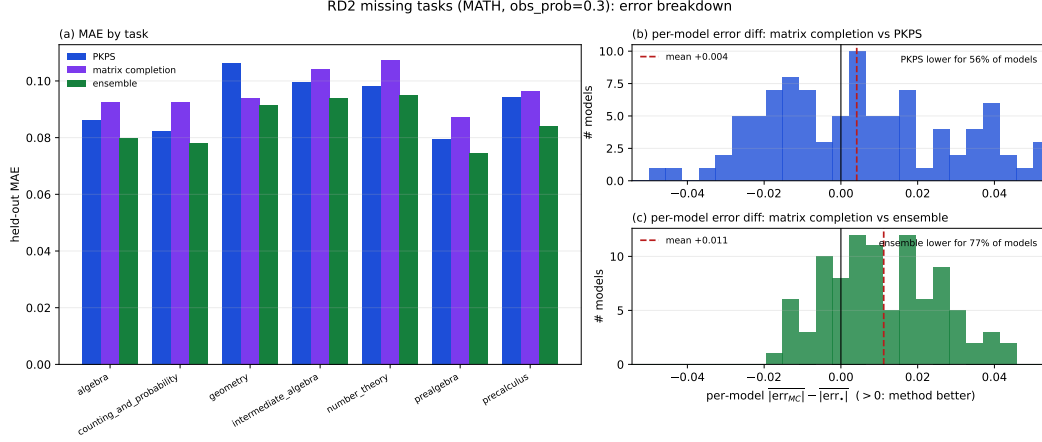


Figure 4: Error breakdown (MATH, obs. prob. 0.3). (a) The ensemble has the lowest MAE on all seven subjects. (b,c) It shifts the whole per-model error distribution down relative to matrix completion (lower for 77% of models), so the gain is broad rather than driven by a few models.

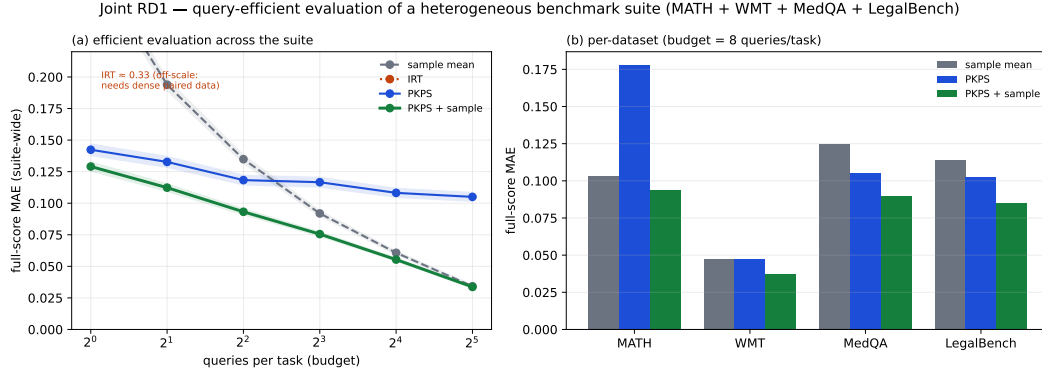


Figure 5: Joint efficient evaluation across the suite (full-score MAE). (a) The PKPS + sample ensemble dominates naive sampling at every query budget; IRT is off-scale. (b) The ensemble is lowest on all four datasets at four queries per task.

alone is weaker because the cross-model perspective discards a model’s own answered queries; the ensemble combines the two. IRT, the standard efficient-benchmarking baseline, needs a dense paired block per task and is off-scale in this sparse, heterogeneous regime.

Benchmark completion (missing tasks). Predicting held-out (model, task) cells of the joint matrix is, by contrast, matrix completion’s regime: capable models tend to be capable across all four domains, so the 18-task matrix is dominated by a single general-ability factor and is highly low-rank. PKPS’s response perspective, which excels at separating closely-related MATH subjects, has little to add once the target is a model’s deviation from its general ability—exactly what the low-rank model already captures (we confirmed this is intrinsic, not an artifact of the joint embedding, across several perspective variants). Matrix completion is therefore the stronger single method here; the PKPS ensemble tracks it and remains the most robust choice, never losing by more than the noise and recovering its margin when models are scarce (Figure 6). Together the two regimes delineate where the response perspective helps: it is decisive for query efficiency and for low-rank-poor, heterogeneous-within-domain completion (MATH), and reduces to the low-rank view as the benchmark matrix becomes rank-one (WMT, the joint suite).

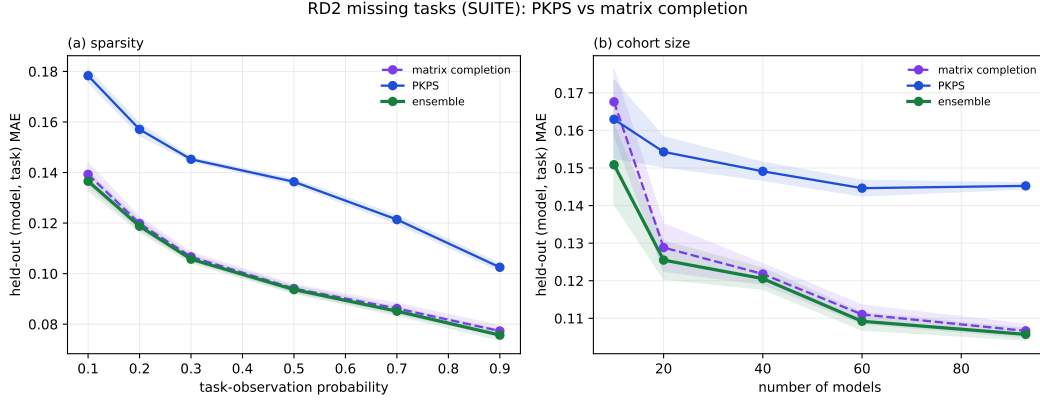


Figure 6: Joint benchmark completion (held-out MAE). The 18-task matrix is low-rank, so matrix completion is strong and PKPS trails; the ensemble (green) tracks matrix completion and is the most robust predictor across sparsity (left) and cohort size (right).

7 Related work

Efficient benchmarking reduces evaluation cost by scoring on a subset of items: item-response theory [Polo et al., 2024, Rasch, 1960] and anchor points [Vivek et al., 2024] select informative items, while DKPS [Anonymous, 2026] predicts from cached responses in a perspective space. These assume a paired/anchored design. **Benchmark completion** treats the score matrix as low-rank and imputes missing entries [Papailiopoulos et al., 2026, Mazumder et al., 2010]; this uses only scores, not responses. PKPS unifies the two settings through a single product-kernel mechanism that consumes unpaired responses, and we show it is complementary to the score-only low-rank view.

8 Conclusion

PKPS extends data-kernel perspective spaces to the sparse, unpaired evaluations that real benchmarks produce, with classical DKPS as its $\sigma \rightarrow 0$ limit and a cheap, cross-validated bandwidth. On a single heterogeneous suite of 18 tasks and 93 models—math, translation, medical QA, and legal classification—a block-diagonal product kernel handles rich-text and single-token responses together and roughly halves the cost of query-efficient evaluation. Our experiments also map *where* the response perspective helps: it is decisive for efficient benchmarking and for completing heterogeneous, low-rank-poor matrices (MATH), and reduces gracefully to the low-rank view as the benchmark matrix approaches rank one (WMT, the joint suite). Across every regime the PKPS–baseline ensemble is the most robust predictor, since the response-embedding and score-only signals are complementary. This makes score prediction practical for organically-assembled leaderboards, where pairing cannot be assumed.

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